

Supplementary figures - paper preview (filled to common column width, 150 dpi)

(si_performance_ranking.png)

PAIRED														
FUD (↓)			LPIPS (↓)			PQ (↑)			DICE (↑)			HED (↓)		
Rank	CT/CS/VAS/NUT	Mean	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ○ ● ●	118.697	1	● ● ● ●	0.319 ± 0.083	1	● ● ○ ●	0.809 ± 0.108	1	● ● ○ ●	0.904 ± 0.107	1	● ● ● ●	0.035 ± 0.023
2	○ ● ● ●	121.582	2	○ ● ● ●	0.327 ± 0.083	2	● ● ● ●	0.808 ± 0.107	2	● ● ● ●	0.903 ± 0.104	2	○ ● ● ●	0.042 ± 0.023
3	● ○ ○ ●	129.243	3	● ○ ● ●	0.330 ± 0.083	3	○ ● ● ●	0.730 ± 0.135	3	○ ● ● ●	0.856 ± 0.130	3	● ○ ● ●	0.044 ± 0.028
...			...			...			...			...		
13	● ○ ○ ○	161.815	13	○ ○ ● ○	0.425 ± 0.068	13	○ ○ ● ●	0.133 ± 0.174	13	○ ○ ● ●	0.277 ± 0.246	13	● ● ○ ○	0.159 ± 0.050
14	○ ● ○ ○	171.584	14	○ ● ○ ○	0.429 ± 0.084	14	○ ○ ○ ●	0.131 ± 0.174	14	○ ○ ○ ●	0.267 ± 0.249	14	● ○ ● ○	0.160 ± 0.051
15	○ ○ ● ○	185.789	15	● ○ ○ ○	0.430 ± 0.083	15	○ ○ ● ○	0.061 ± 0.115	15	○ ○ ● ○	0.143 ± 0.179	15	○ ○ ● ○	0.166 ± 0.051

UNPAIRED														
FUD (↓)			LPIPS (↓)			PQ (↑)			DICE (↑)			HED (↓)		
Rank	CT/CS/VAS/NUT	Mean	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD	Rank	CT/CS/VAS/NUT	Mean ± SD
1	○ ○ ○ ●	136.401	1	● ● ● ●	0.478 ± 0.121	1	● ● ○ ●	0.765 ± 0.167	1	● ● ○ ●	0.847 ± 0.193	1	● ● ● ●	0.044 ± 0.031
2	○ ○ ● ●	139.489	2	○ ● ● ●	0.491 ± 0.119	2	● ● ● ●	0.757 ± 0.164	2	● ● ● ●	0.839 ± 0.195	2	○ ● ● ●	0.046 ± 0.026
3	● ○ ○ ●	142.337	3	● ○ ● ●	0.492 ± 0.120	3	● ● ○ ○	0.681 ± 0.159	3	● ● ○ ○	0.827 ± 0.153	3	● ○ ● ●	0.049 ± 0.030
...			...			...			...			...		
13	● ● ○ ○	180.869	13	● ○ ○ ○	0.547 ± 0.113	13	○ ○ ○ ●	0.140 ± 0.115	13	○ ○ ○ ●	0.308 ± 0.175	13	● ● ● ○	0.160 ± 0.053
14	● ● ● ○	183.658	14	○ ○ ○ ●	0.549 ± 0.102	14	○ ○ ● ●	0.134 ± 0.110	14	○ ○ ● ●	0.298 ± 0.171	14	● ○ ● ○	0.162 ± 0.055
15	○ ○ ● ○	189.310	15	○ ○ ● ○	0.564 ± 0.104	15	○ ○ ● ○	0.064 ± 0.083	15	○ ○ ● ○	0.171 ± 0.153	15	○ ○ ● ○	0.172 ± 0.054

Figure S1. Top- and bottom-ranked channel-group conditions per metric, for paired and unpaired inference. Top-3 and bottom-3 ranked conditions per metric — FUD (↓), LPIPS (↓), PQ (↑), DICE (↑), HED (↓) — for paired (upper) and unpaired (lower) inference. The dot column encodes active channel groups in the order CT, CS, VA, NU, UNI (filled = included): CT (Cell types) groups the healthy / cancer / immune density maps; CS (Cell state) groups the proliferative / nonproliferative / dead density maps; VA (Vasculature) is the vasculature density channel; NU (Nutrient) groups the oxygen and glucose channels; UNI is the UNI-2h H&E reference embedding (paired-only). The mean and per-tile mean ± SD are listed per ranked condition.

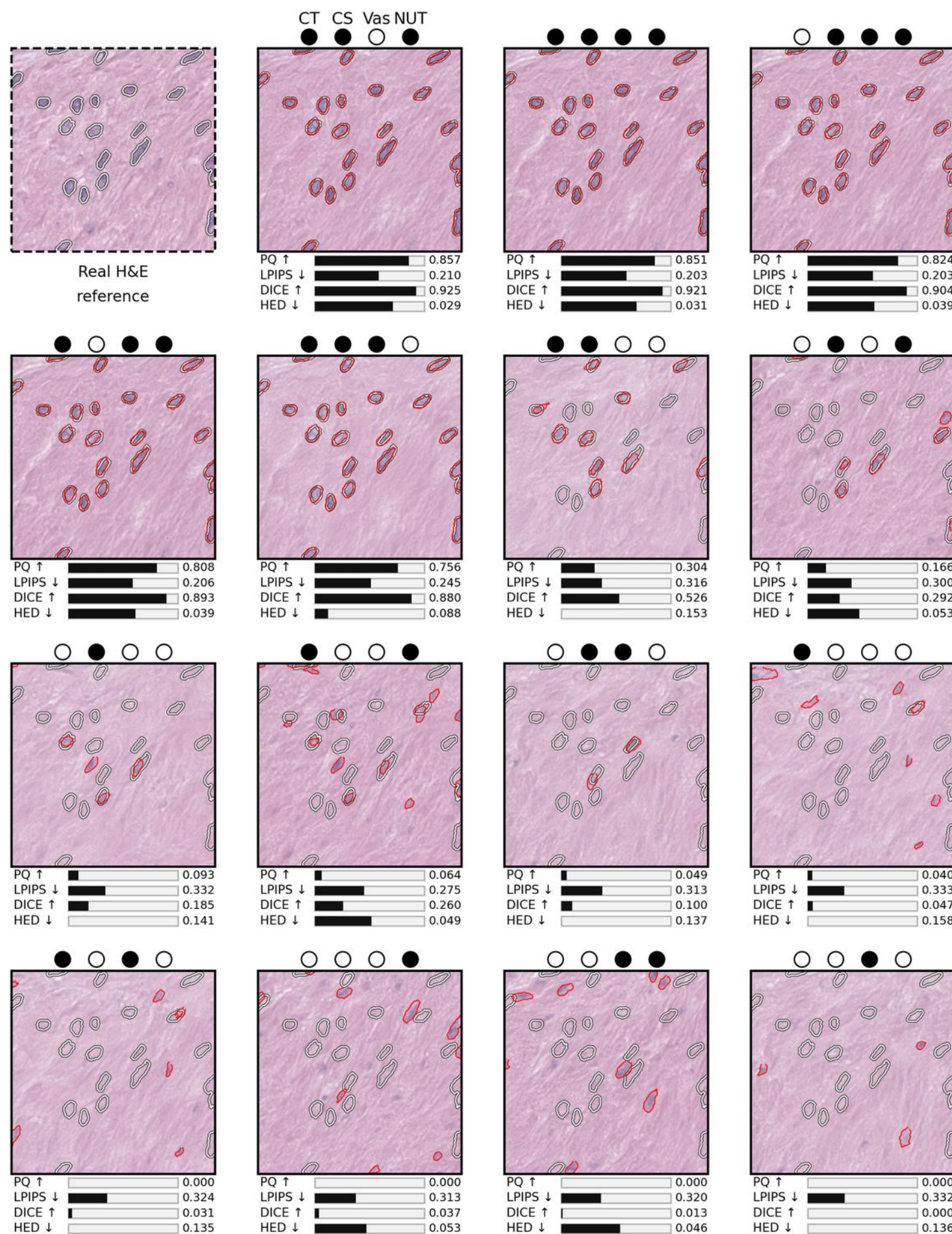
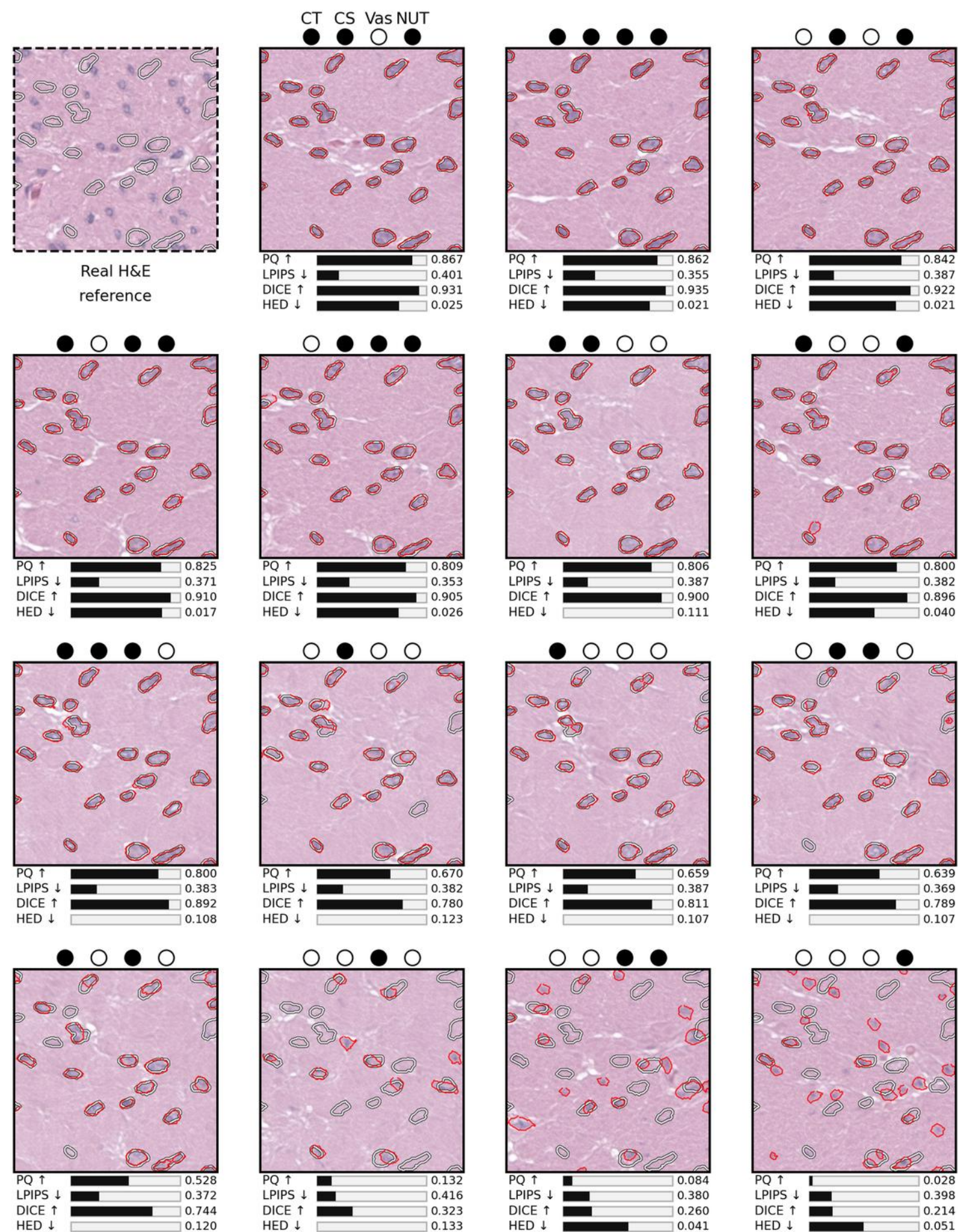
A Paired ablation grid**B Unpaired ablation grid**

Figure S2. Generated H&E patches under channel-group ablations illustrate ControlNet output for paired and unpaired inference. (A) Paired-inference results for one representative tile. **(B)** Unpaired-inference results for one representative tile. Each block is a 4 × 4 grid; the first cell (gray dashed border, top-left) is the real H&E reference for that tile, and the remaining 15 cells are generated patches sorted by Panoptic Quality (PQ; higher is better) from best to worst across channel-group on/off combinations. The 4-dot header above each cell encodes active channel groups in the order CT, CS, VA, NU (Cell types, Cell state, Vasculature, Nutrient); a filled dot indicates the group is included. Per-cell bar plots report LPIPS (lower is better), PQ (higher is better), DICE (higher is better), and HED (lower is better) for that single patch, with the numeric value printed beside each bar.

(si_a1a2_qualitative_tiles.png)

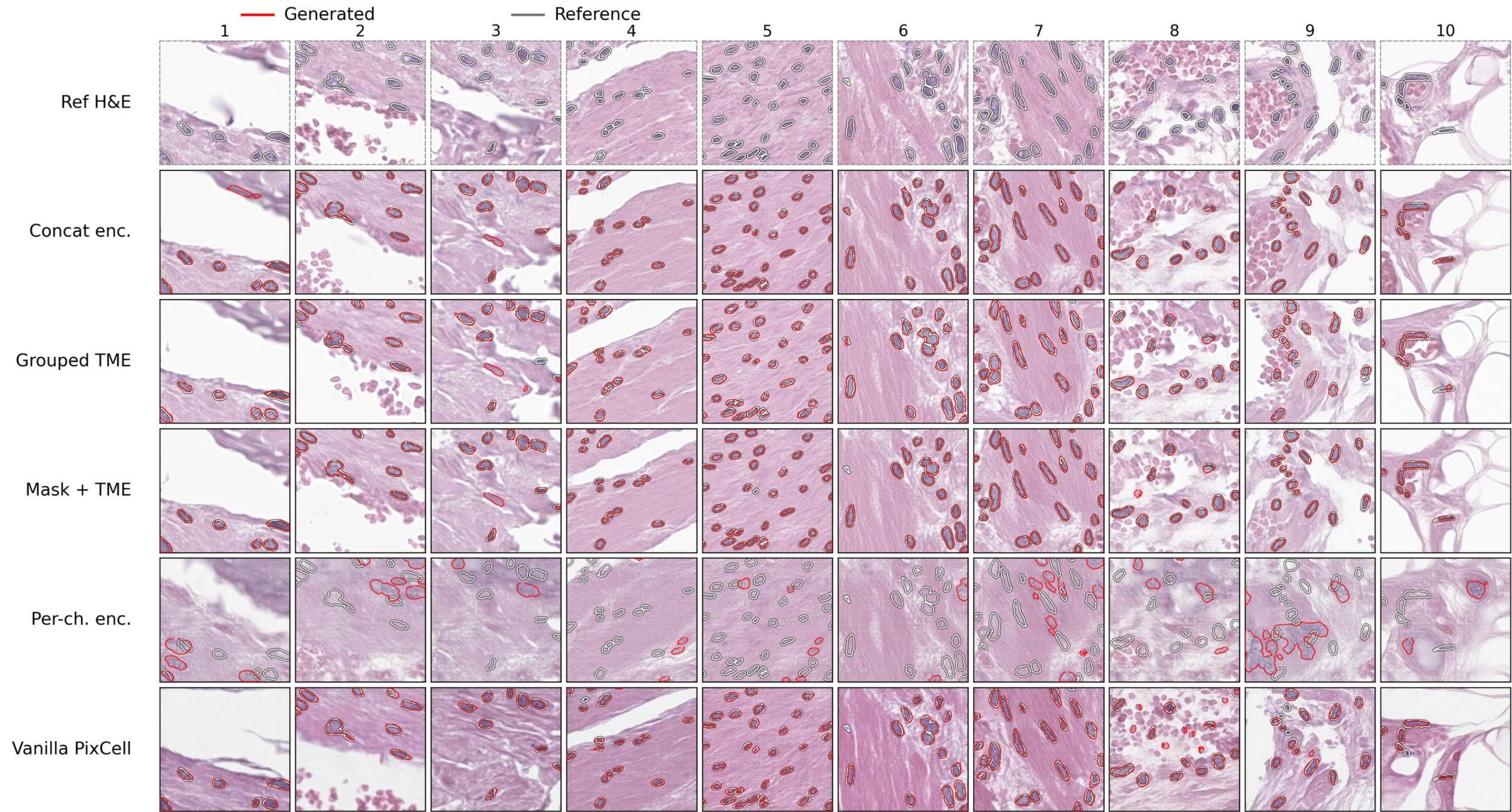


Figure S3. Conditioning-architecture ablation: qualitative H&E for each TME-injection variant. Qualitative tiles across 10 columns for the conditioning-architecture variants — Grouped TME only, Concat TME encoder, Per-channel TME encoders, additive Mask + TME, and an off-the-shelf Vanilla PixCell ControlNet baseline. The top row is the real reference H&E; each lower row is one variant's generated H&E, with overlaid contours marking segmented nuclei (red = generated, gray = reference) so segmentation fidelity can be compared per tile. The full per-variant metrics table (FUD ↓, DICE ↑, PQ ↑, LPIPS ↓, HED ↓ and trainable parameter count) is provided separately as `individual/si_a1_a2/SI_A1_A2_section2_metrics.png`.

(si_channel_decodability_impact.png)

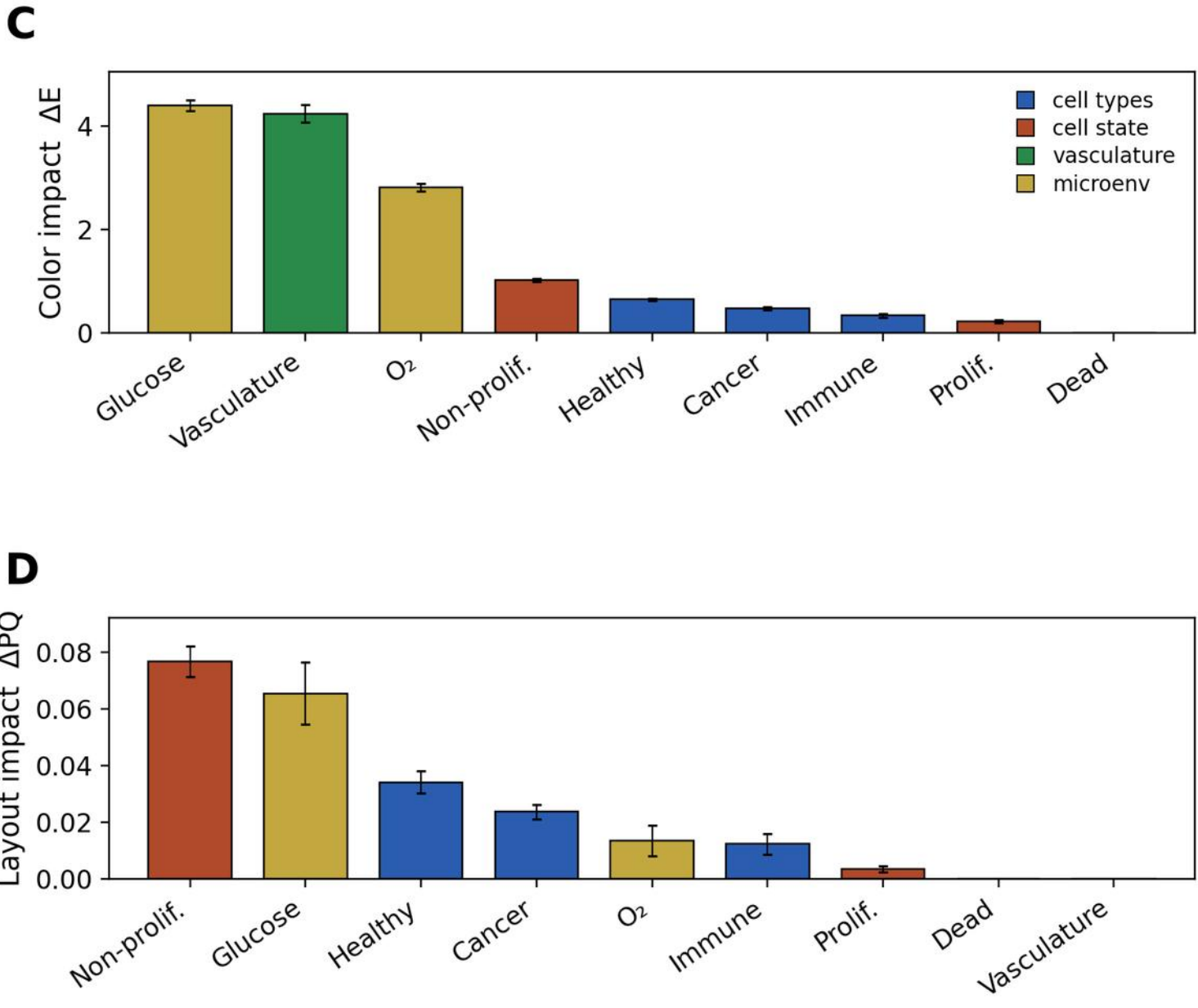
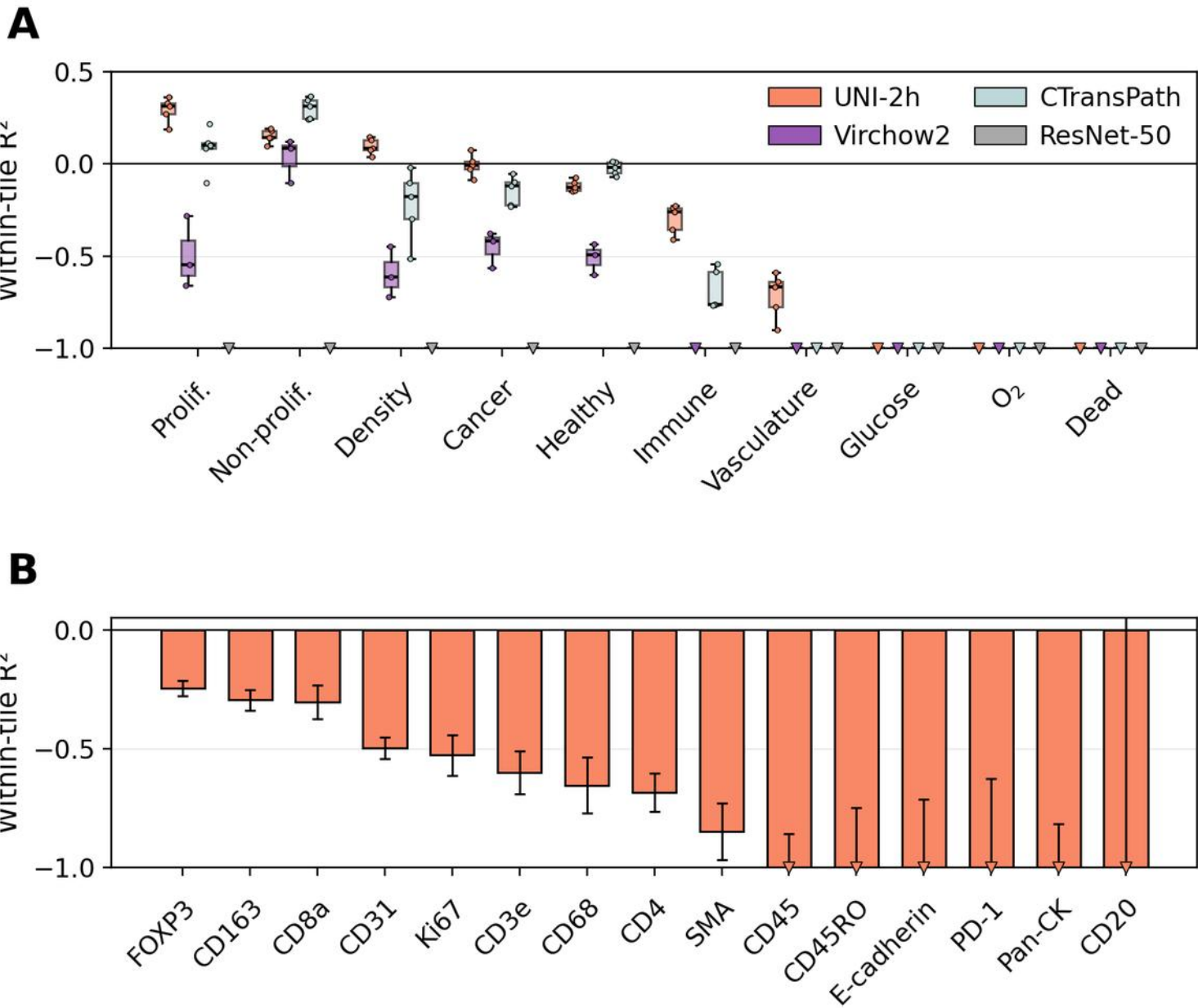


Figure S4. H&E decodability of TME channels/markers and the ranked per-channel generative impact behind Fig 4B. (A) Within-tile decoding R^2 for each aggregate TME channel (Prolif, Non-prolif, Density, Cancer, Healthy, Immune, Vasculature, Glucose, O_2 , Dead) across four frozen H&E encoders (UNI-2h, Virchow2, CTransPath, ResNet-50), one boxplot per encoder per channel. **(B)** Within-tile decoding R^2 for individual MX marker intensities using the best encoder (UNI-2h), one bar per marker ordered by descending R^2 (error bars = CV-fold standard error). Panels A-B are the decodability axis (x) of Fig 4B. **(C)** Per-channel color impact, mean CIEDE2000 ΔE between full-conditioning and leave-one-out generations, measured over 300 tiles from the sub-channel ablation (mean $\pm$ SEM; bars colored by group). **(D)** Per-channel layout impact, drop in Panoptic Quality (ΔPQ) of CellViT-segmented nuclei relative to full conditioning. Channels in C-D are ranked descending; these are the impact axes (y) of Fig 4B.